Using the BL5A GUI

This is the operator's guide to the window: what each part shows, what each button sends, and the two experiments from start to end. To build the programs and set them up at the beamline, see install.md.

The screenshots were taken against the simulation (`bl5a-sim`), so the ring current, shutter and temperature fields read "—" in them.

Before you start

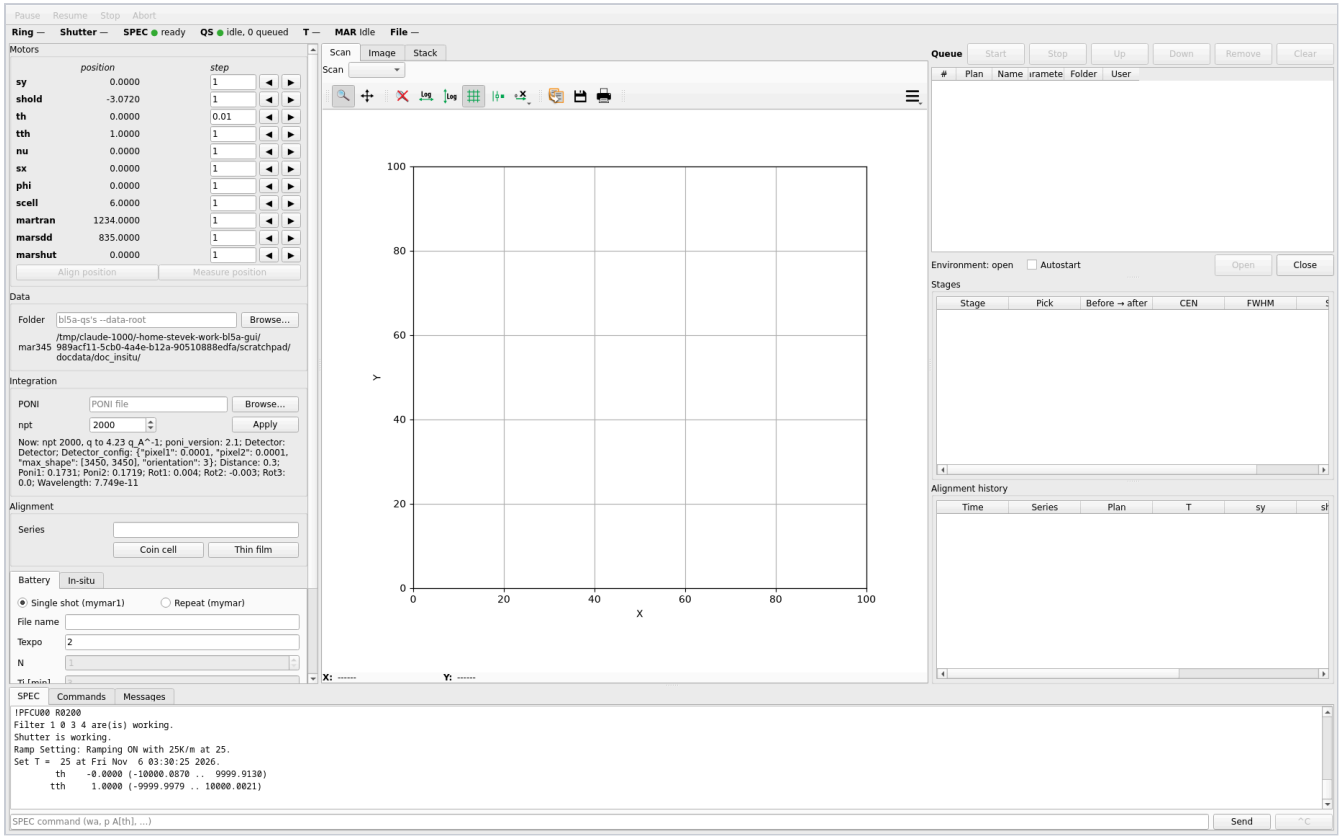
The GUI is a client. On its host, user systemd keeps spec-ioc, `bl5a-qs` (the queue server, which runs the plans) and `bl5a-integrator` (which integrates each frame) running (install.md, section 5); the beamline runs the mar345 IOC. You start SPEC in server mode, then the GUI with the **BL5A** shortcut on the desktop or in the applications menu. `systemctl --user status 'bl5a*'` shows whether the services run.

In a terminal, `deploy/bl5a-gui` starts the same window and takes these options:

Option	Default	What it is
`--qs-control`	`tcp://localhost:60715`	bl5a-qs control socket
`--qs-documents`	`tcp://localhost:60725`	bl5a-qs document stream
`--integrator-rep`	`tcp://localhost:60735`	integrator requests
`--integrator-pub`	`tcp://localhost:60745`	integrator new rows and errors
`--user`	`bl5a`	the name in the queue's User column

You can close the GUI at any time. The queue keeps running in bl5a-qs and the stacks stay with the integrator. A new window starts with empty Scan, Stages, Alignment history, Commands and Messages panes: they are filled from what the plans report while the window is open. The Stack tab asks the integrator, so it shows every stack it holds.

The window



The window at rest

Area	What is in it
Top	the run buttons (Pause, Resume, Stop, Abort) and the status line
Left	motors, the Data and Integration boxes, the alignment buttons, the Battery and In-situ forms
Centre	the Scan, Image and Stack tabs
Right	the queue, the stages of the running alignment, the alignment history
Bottom	the SPEC console, the Commands the plans sent, the GUI's Messages

The panes are split by handles you can drag.

Who drives SPEC

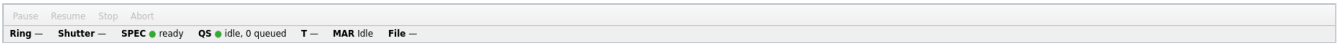
SPEC runs one command at a time. While the queue runs a plan, the plan owns SPEC, so the GUI greys out everything that would send SPEC a command of its own: the motor tweak buttons, their step fields and the SPEC console's entry.

They are live only when all of these hold:

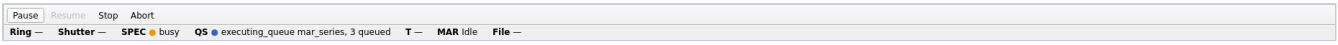
- bl5a-qs is idle and runs nothing;
- spec-ioc reports SPEC connected (`BL5A:SPEC:Connected`);
- SPEC is ready (`BL5A:SPEC:Busy`);
- no console command is running (`BL5A:SPEC:Cmd:Busy`).

The queue itself can be edited at any time.

Status line



The status line at rest



The status line while a mar_series runs

Field	Shows
Ring	the ring current, mA (`BL5A:userCalcOut1.VAL`)
Shutter	the front-end shutter; its PV is not configured yet, so "—"
SPEC	● ready (green), ● busy (orange), ● disconnected (red); "—" when spec-ioc does not answer
QS	the queue server's state, the running plan, how many items wait; "environment closed" and "stopping after the frame" when they apply; ● not answering (red) when bl5a-qs does not reply
T	the temperature as readback → setpoint, and the ramp; "—" until the temperature PVs are configured
MAR	the mar345 detector state (`DetectorState_RBV`)
File	the full path of the last frame; you can select and copy it

The QS dot is green when idle, orange when paused and blue otherwise.

Run buttons

Button	Live when	What it does
Pause	a plan runs	pauses at once; a moving motor is stopped
Resume	paused	goes on
Stop	a plan runs or is paused	ends the plan, recorded as a success
Abort	a plan runs or is paused	asks first, then ends the plan, recorded as aborted

However a plan ends, bl5a-qs then interrupts a SPEC command still running, stops a count still running and closes the mar345 shutter.

The plans do not replay a SPEC macro that was cut in half. After **Pause**, **Resume** fails the plan with "plan interrupted: a paused plan cannot resume, abort it"; end it with **Stop** or **Abort**. To end a battery series cleanly, use **Stop after frame** on the Battery form instead: that pause lands between frames, where the series can go on.

A motor's stop reaches SPEC as `abort_all`, which stops every motor SPEC is moving.

The queue's own **Stop** is different: it lets the running item finish and then stops the queue.

Motors



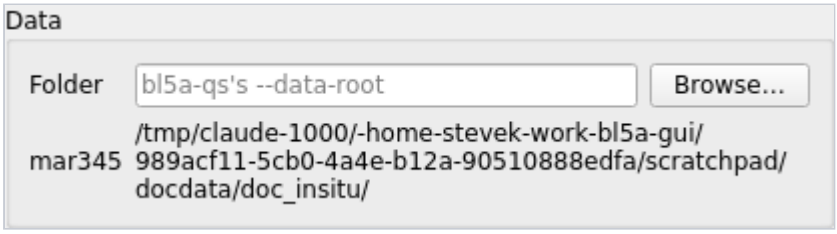
The motors panel

One row per spec-ioc motor: `sy shold th tth nu sx phi scell martran marsdd marshut`.

- **position** is the readback, with the record's precision and unit.
- An orange dot after it means the motor is moving.
- **step** is the tweak step (the motor record's `TWV`). Type a positive number and leave the field (Tab or click elsewhere) to set it.
- ◀ moves the motor by −step, ▶ by +step. spec-ioc turns each into a SPEC move.

Align position and **Measure position** stay disabled until BL5A gives these positions.

Data



The Data box

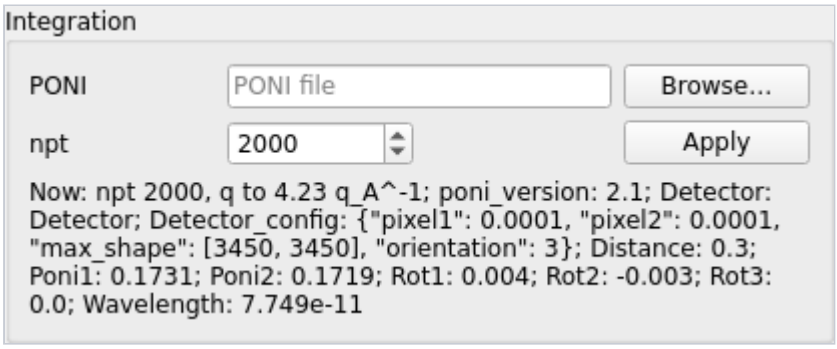
Everything an item records goes in one folder, `<data root>/<series>/`: its frames, its scans as the SPEC file `<series>.spec` and its integrated stacks. bl5a-qs makes the folder when the item starts. The series is the one typed in the form, or the plan's name when there is none, so the items of one series share a folder. Type the same series in **Alignment** and on the **Battery** form and a cell's alignment scans and frames land together:

```
<data root>/
  cell7/
    cell7.spec              the align_coin scans
    cell7_0001.mar3450 …    the frames
    cell7.h5                the stack
  film_A/
    film_A.spec             each in-situ row's alignment scans
    filmA_30_0001.mar3450 …
    film_A.h5
```

- **Folder** is the data root of the items queued from now on; empty means bl5a-qs's `--data-root`. **Browse…** picks it, and the GUI remembers it. It must be an absolute path without spaces that is the same on the bl5a-qs, mar345 IOC and integrator hosts (see install.md).
- Each queued item carries the folder as it was when queued, so a change does not move items already in the queue. The queue's **Folder** column shows where each item goes.
- **mar345** is the folder the mar345 IOC writes to now (`FilePath_RBV`); bl5a-qs sets it before each frame. A red "missing" after it means the IOC does not see that folder (`FilePathExists_RBV` reads No).

A series is one folder name without spaces, `/` or control characters, and not `.` or `..`; the whole folder, `<data root>/<series>`, is at most 200 bytes. The mar345 detector server takes a frame's path as one word, so a space would split it. The forms refuse any other series, and queueing refuses a **Folder** with a space or a path that is too long. In the stack's file name, a character other than a letter, digit, `_`, `-` or `.` is written `%XX`: series `LiFePO4(1C)` gives `LiFePO4%281C%29.h5`. An item whose data root does not exist fails when it starts, with "data folder: … the data root … does not exist".

Integration



The Integration box

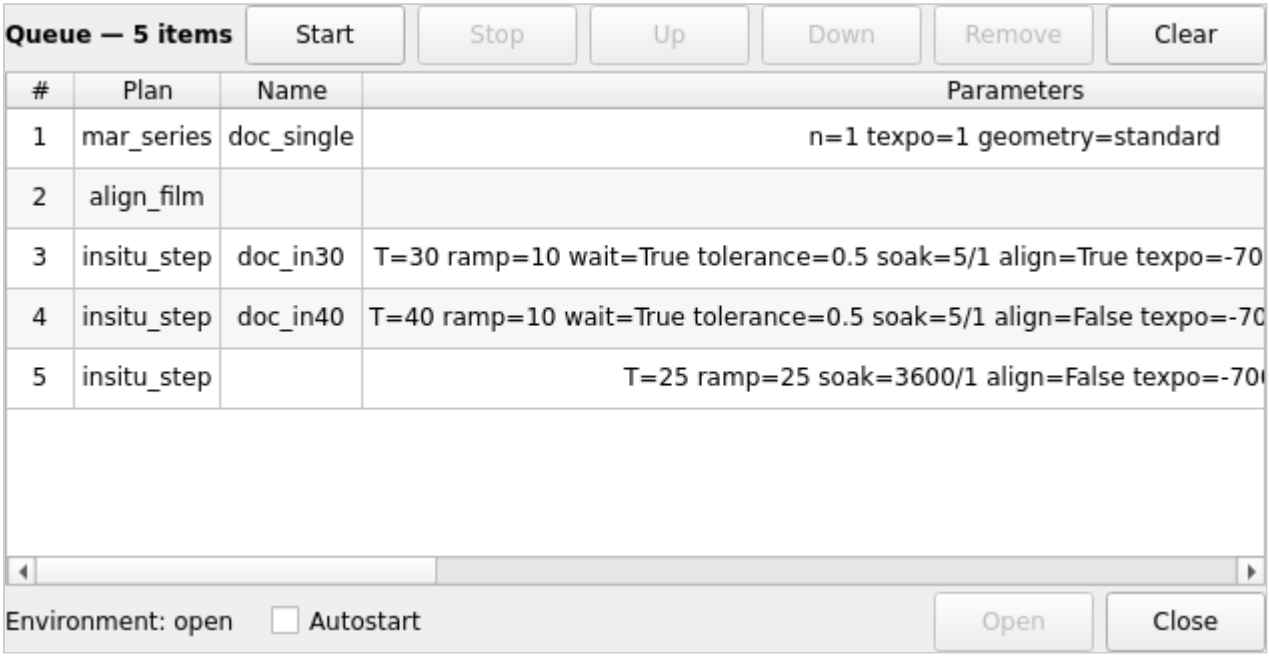
The text under the buttons is what the integrator integrates with now: npt, the q range and the PONI.

Control	What it is
PONI	a pyFAI PONI file for the full 345 mm plate; **Browse…** picks it
npt	the bins from q = 0 to the plate's far rim (default 2000)
Apply	reads the file and sends its text and npt to the integrator

The integrator checks them first: a PONI it cannot use is refused, and the box adds "Refused:" and the reason. An accepted one applies to the frames after those already waiting to be integrated; until then the box shows it as "Next, after the queued frames". When it takes effect, Messages says "integrator: npt …, q to … from now on" and the Image tab's rings move to the new PONI.

A new npt, or a PONI that moves the q range, starts a new stack: the next frame of `cell7` goes to `cell7~2.h5`, and `cell7.h5` keeps its q axis. The Stack tab lists both. The integrator keeps the setting until it restarts, and then starts again from its `--poni` and `--npt`.

Queue



The queue with five items

The top row, marked ▶ on a green background, is the running item; the rest wait in order.

Column	Shows
#	the place in the queue
Plan	`mar_series`, `align_coin`, `align_film` or `insitu_step`
Name	the frame file name, when the plan takes frames
Parameters	the item's arguments, for example `T=30 ramp=10 wait=True tolerance=0.5 soak=5/1 align=True texpo=-70000 th_list=-0.05/0.11/0.13/0.15`
Folder	where the item's data goes: `<data root>/<series>`, or the series alone when the item goes to bl5a-qs's `--data-root`
User	the `--user` that queued it

Button	What it does
Start	runs the queue from its first item (only with items and nothing running)
Stop	stops the queue after the running item; that item is not interrupted
Up, Down	moves the selected item one place
Remove	takes the selected items out (also the Delete key); not the running item

Button	What it does
Clear	empties the queue after asking; the running item goes on

The row under the list belongs to the queue server:

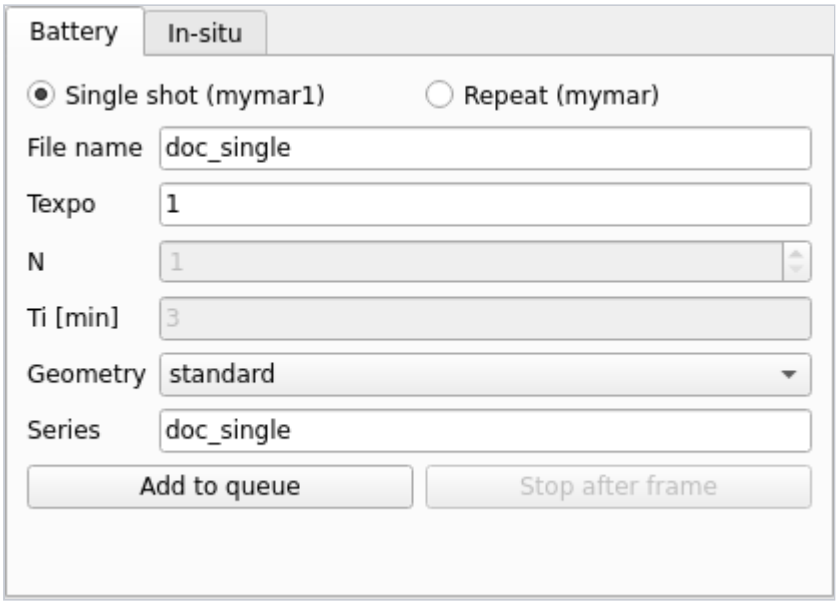
- **Environment: open / closed / not known.** Nothing runs while it is closed. **Open** opens it. **Close** is offered only while idle.
- **Autostart.** When ticked, what is added to the queue runs without a Start. The server unticks it when the queue ends in any way other than running dry: a failed or aborted plan, or a Stop. Tick it again here.

A click on a form's queue button sends one request. The rows of the In-situ table go in as one request, in table order, so nothing can slip in between them.

bl5a-qs checks an item's arguments only when the item runs. The forms check the same rules before they queue, so a bad value is caught at the form and not halfway through the night.

Battery measurements

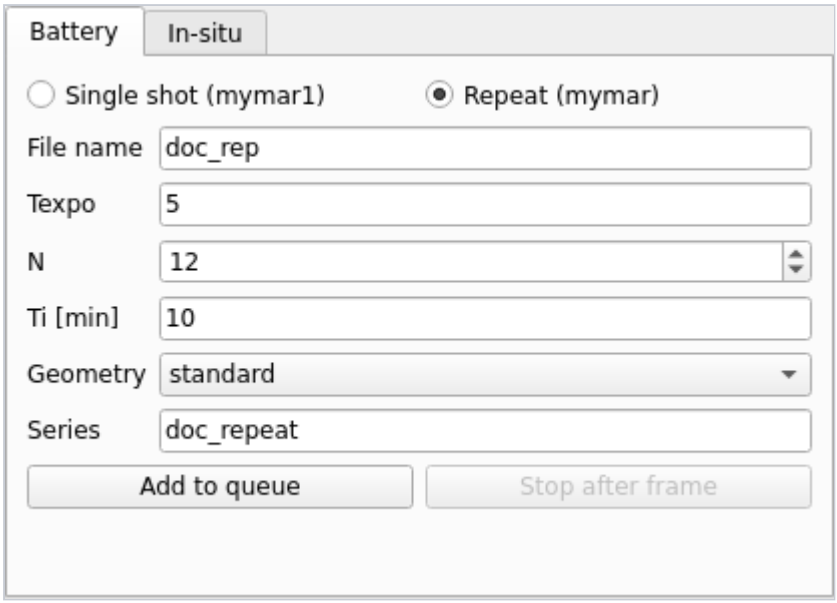
The Battery form queues `mar_series`: SPEC's `mymar1` (one frame) or `mymar` (a frame every Ti minutes).



Single shot

Field	Rule	Notes
Single shot / Repeat		Single shot is `mymar1`; Repeat is `mymar`
File name	1 to 27 characters, no spaces or quotes	the mar345 file name
Texpo	seconds 1 to 20, or a negative number of monitor counts	default 2
N	whole number, at least 1	Repeat only: the number of frames
Ti [min]	more than 2.5	Repeat only: from one frame's start to the next

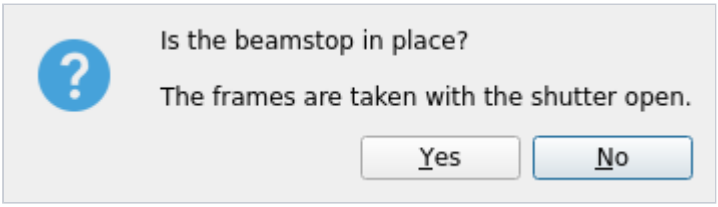
Field	Rule	Notes
Geometry	`standard`, `high_q_asymm`, `low_q_symmetric`	mymar's mar345 positions, below
Series	a folder name (Data)	the data folder and stack the frames go to; empty means the file name



Repeat: twelve frames, ten minutes apart

Geometry	`mar_ip7` arguments
standard	`-85 435 100 330 50 1.2 -6.8`
high_q_asymm	`-85 435 110 330 80 1.2 -6.8`
low_q_symmetric	`160 435 36 345 54 1.2 -6.8`

Add to queue checks the fields first. A bad value opens a "Battery" box that names the field and the rule. Then the GUI asks:



The beamstop question

Yes queues the item; **No** queues nothing.

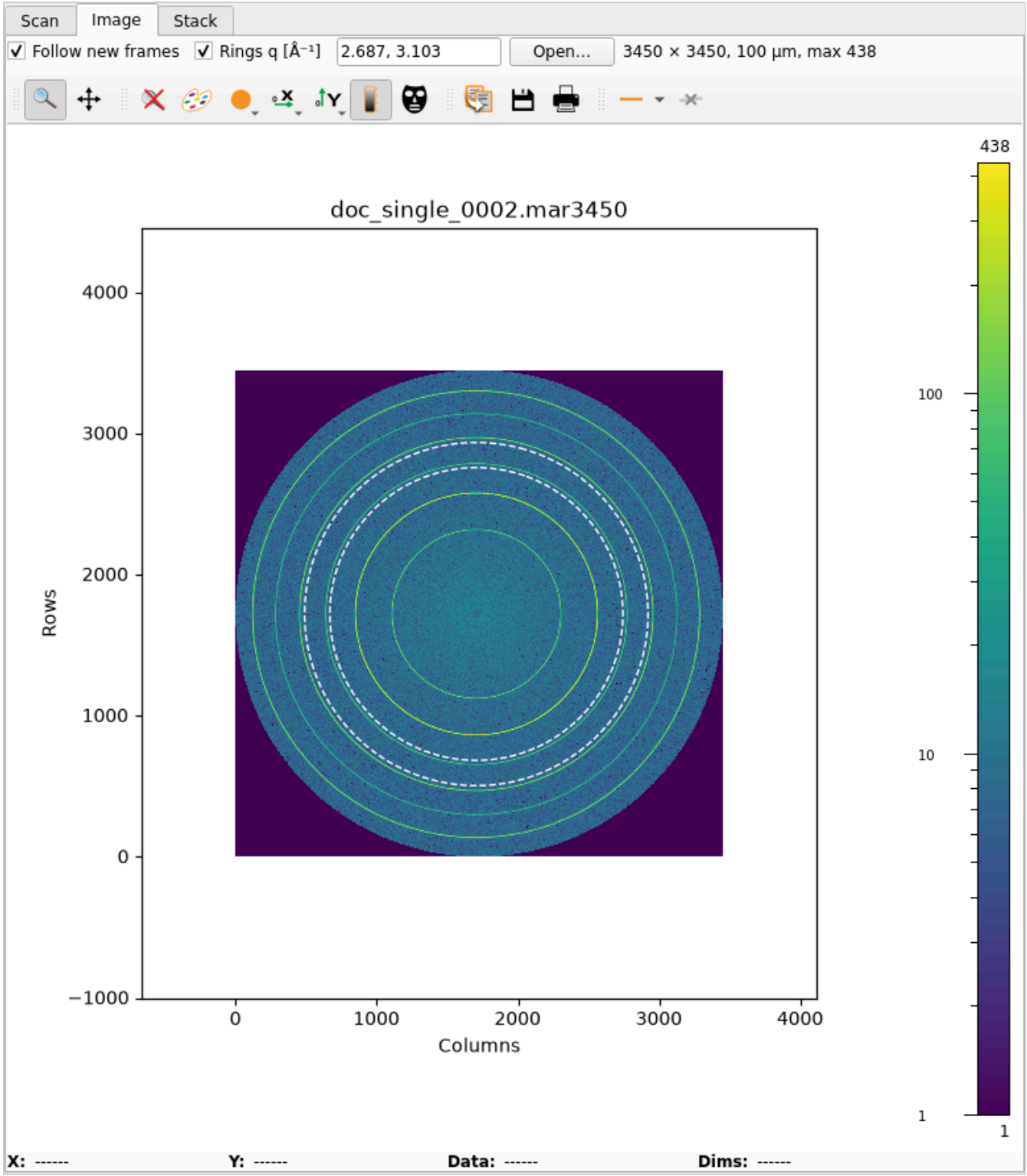
Move tth to 80° or more first. `mar_series` does not move tth, and `mar_ip7` refuses the upward geometry below 80° ("Two Theta Angle is too low for UPWARD Geometry."). In the simulation the plan then fails with that message; this has not been tried against the real SPEC. While the queue is idle you can type `umv tth 81` in the SPEC console. bl5a-sim does not model `umv`; there, set tth's step and click ▶ instead.

What a `mar_series` does, in order:

1. `mar345closeshutter`, `scell` to 0, `counters 2 3`, `fi 0 1 0 0`, `ic_range 4`, a 2 s wait, `mar_ip7 <geometry>`, `scell` to 0.

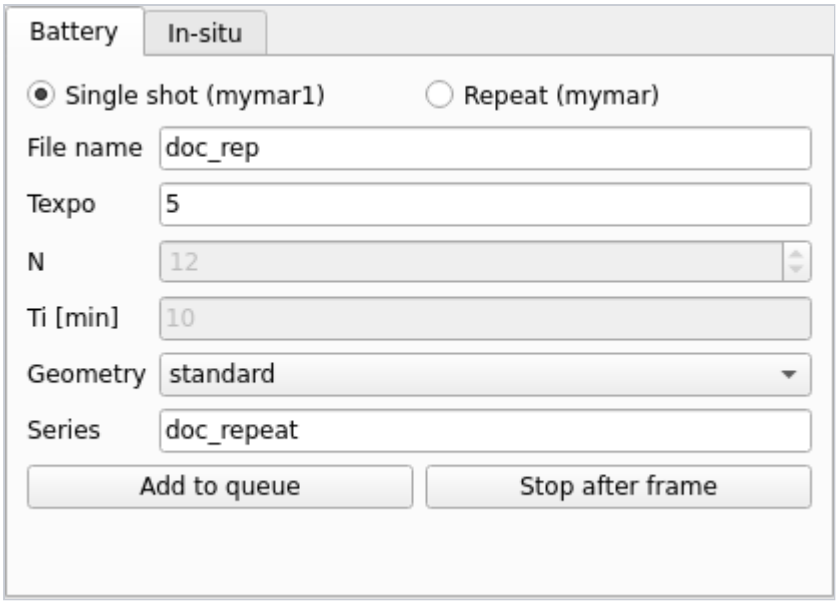
2. For each frame: the file name goes to the mar345 IOC, `mar345openshutter`, the count (in pieces of at most 429 s, or one monitor count), `mar345closeshutter`, `exp_condition`, the plate readout through the IOC, `mar345readout`.
3. Frame *i* starts *i*·Ti after the first frame. Between frames the plan only waits.

The mar345 writes `<name>_<number>.mar3450` in the item's data folder. The number is the IOC's file counter, which runs on from one name to the next.



A single shot on the Image tab

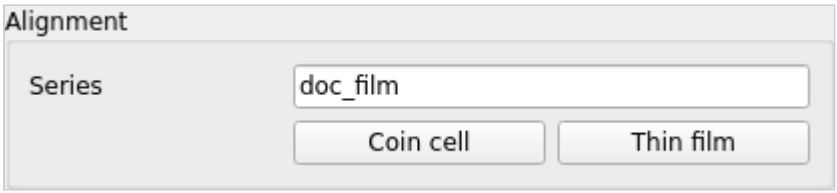
Each frame appears on the Image tab once it is read out, and is added to its series on the Stack tab once the integrator has integrated it.



While a series runs

Stop after frame is live while a `mar_series` runs. It lets the frame being taken finish, then ends the series: it asks for a pause that takes effect between frames, and sends Stop once the plan has paused. The status line says "stopping after the frame" in the meantime.

Alignment



The alignment buttons

Series groups the runs in the alignment history, is stored with the run and names the data folder its scans go to. Empty means each run stands on its own, and its scans go to the folder `align_coin` or `align_film`.

Coin cell queues `align_coin`, SPEC's `coin_align`:

1. `marshut` to 0, `mon1_setup 1050 100 300 3.0 1.0`, `setkappa 50`, `counters 2 5`, `fi 1 0 0 0`, `fil 0 0 0 0`.
2. A fly scan of sy from −2.5 to 2.5 on ic1; sy goes to pl_COM.
3. A fly scan of shold from −2 to 2; shold goes to pl_COM.
4. `mar345closeshutter`, the plate is erased, `marshut` to 340.
5. The positions of sx, sy, shold, th and phi are recorded.

`coin_align` ends with `mar_ip` without arguments, which only prints its usage in SPEC, so the plan does not send it.

Thin film queues `align_film`, SPEC's `align_in.mac`. It sets the slits, attenuators and counters, moves to the direct beam (`uan 1 0`, `all6 0 0 89.7 0 0 0.0`, `shold` to −3) and then runs these stages. Each stage scans, moves the motor to the picked statistic and, where a value is given, redefines that position with `set`.

Stage	Pick	Then
tth −0.04 to 0.04	CEN	`set tth 0`
nu −0.1 to 0.1	CEN	`set nu 0`
shold −3 to −1.5, fly	EDGE	
th −1 to 1 relative, fly	pl_xMAX	`set th 0`
shold 0.3 to −0.3 relative, fly	EDGE	
th −2 to 2 relative, fly	CEN	
shold −0.3 to 0.3 relative	EDGE	
th −0.6 to 0.6 relative	CEN	`set th 0`
shold −0.15 to 0.15 relative	EDGE	
`uan 1.2 0.6`, `atte 4`, then th −0.2 to 0.2 relative, 120 intervals	pl_xMAX	`set th 0.6`

It ends with `mar345closeshutter` and an erase of the plate.

EDGE is where a shold scan falls through half its maximum, the height at which the sample takes half the beam. `align_in.mac` moves to CEN, which for an edge lies halfway between that point and the maximum, and on the open-beam plateau the maximum is wherever the noise puts it: over four scans of one edge on the beamline CEN moved by 0.24 while the half point stayed within 0.007.

Before a stage moves

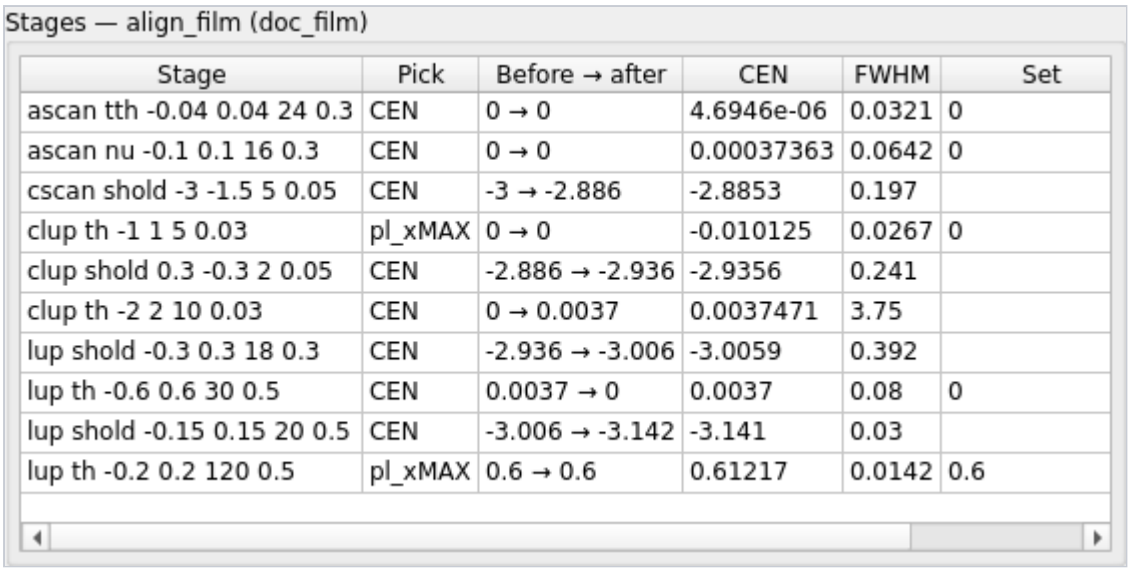
A point more than 3σ from the median of itself and the two points either side of it is replaced by that median, and so is a point that is not a number. σ is that of the five points, but at least the square root of their median and the noise of the whole scan. A cscan's second sample counts about 1.7 times as long as the others and is often replaced this way; left in, it can be the maximum CEN and pl_xMAX follow. The statistics are those of the filtered points.

The stage then puts the motor back where it was before the scan and ends the plan, which fails with the reason Messages shows, when

- the scan shows nothing: averaged into 25 bins, it changes by less than 10% of its largest bin, or by less than 5 times the noise of a bin;
- a peak's maximum is the scan's first or last point, or both of its neighbours are at or below half of it (a peak one point wide);
- an edge never falls to half its maximum, or falls to half on both sides.

A peak that never falls to half inside the scan, such as a rocking curve wider than a th scan, still goes to CEN, which is then the maximum.

Stages


Stages — align_film (doc_film)

Stage	Pick	Before → after	CEN	FWHM	Set
ascan tth -0.04 0.04 24 0.3	CEN	0 → 0	4.6946e-06	0.0321	0
ascan nu -0.1 0.1 16 0.3	CEN	0 → 0	0.00037363	0.0642	0
cscan shold -3 -1.5 5 0.05	CEN	-3 → -2.886	-2.8853	0.197	
clup th -1 1 5 0.03	pl_xMAX	0 → 0	-0.010125	0.0267	0
clup shold 0.3 -0.3 2 0.05	CEN	-2.886 → -2.936	-2.9356	0.241	
clup th -2 2 10 0.03	CEN	0 → 0.0037	0.0037471	3.75	
lup shold -0.3 0.3 18 0.3	CEN	-2.936 → -3.006	-3.0059	0.392	
lup th -0.6 0.6 30 0.5	CEN	0.0037 → 0	0.0037	0.08	0
lup shold -0.15 0.15 20 0.5	CEN	-3.006 → -3.142	-3.141	0.03	
lup th -0.2 0.2 120 0.5	pl_xMAX	0.6 → 0.6	0.61217	0.0142	0.6

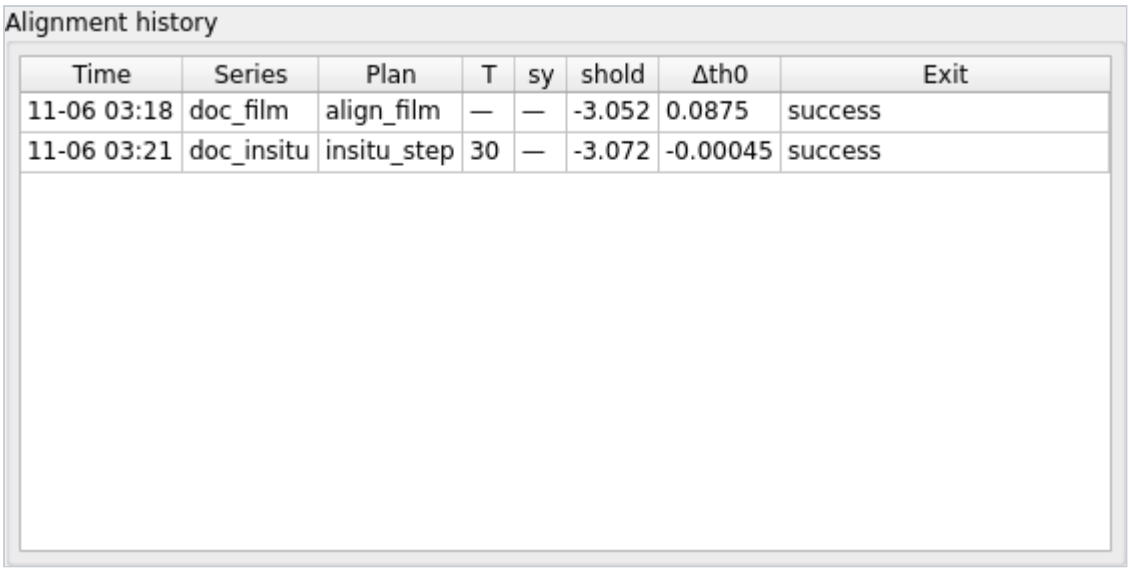
The stages of an align_film run

The Stages table shows each stage of the alignment running now. Its title names the plan and series. It is cleared when the next `align_coin`, `align_film` or `insitu_step` starts, including an `insitu_step` row without alignment.

Column	Shows	Tooltip
Stage	the scan, as a SPEC command line	
Pick	the statistic the motor was moved to: CEN, pl_COM, pl_xMAX or EDGE	the target position
Before → after	the motor before the scan, and after the move and any `set`	
CEN	the centre at half maximum	COM and the position of the maximum
FWHM	the full width at half maximum	
Set	the value given to `set`, empty when the stage did not set	how far the user offset moved

The statistics are splot's (`CEN`, `pl_COM`, `pl_xMAX`), the ones SPEC's macros read, of the points after the outlier filter (see Before a stage moves).

Alignment history



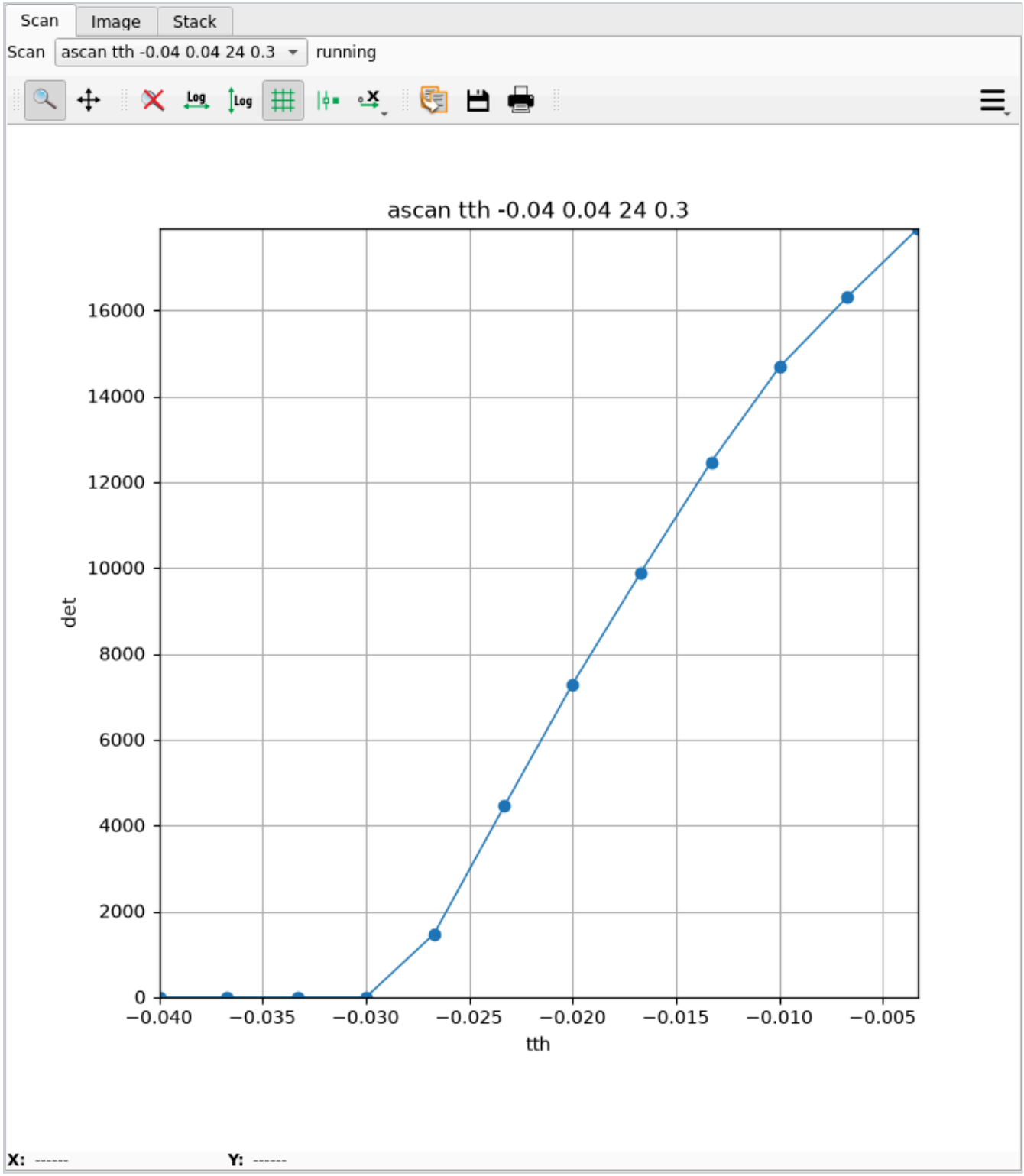
The alignment history

One row per finished run that carried alignment stages (or, for the coin cell, its recorded positions):

Column	Shows
Time	when the run started
Series	the series it was queued in
Plan	`align_coin`, `align_film` or `insitu_step`
T	the `insitu_step` setpoint, °C
sy, shold	where the run left them
Δth0	how far this run moved th's zero
Exit	`success`, `fail` or `abort`; its tooltip gives the reason a run did not succeed

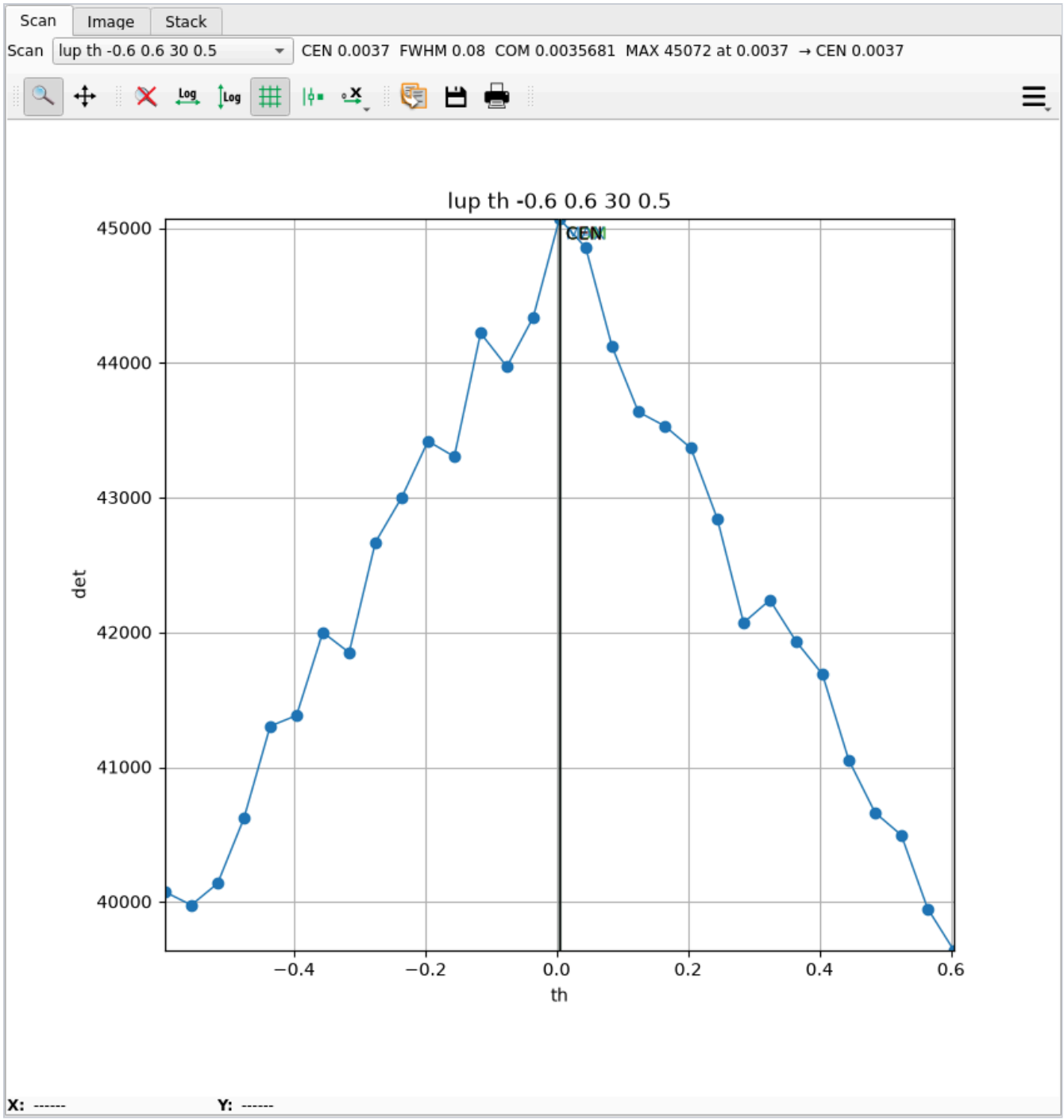
Δth0 is the sum of this run's `set th` corrections, not th0 itself: SPEC's dial position is not published, so the absolute offset is not known to the GUI.

Scan tab



A scan being drawn

A scan is drawn point by point as it runs, at most five times a second. A fly scan runs in SPEC, which keeps its sampling timing; each of its points is drawn as SPEC takes it (spec-ioc passes on SPEC's `SCAN_PT` events), and is the counts SPEC took in one sample, at the middle of the motor's travel over it. When SPEC finishes, its full table of the scan adds any point not seen yet; a table that does not begin with the points already drawn fails the scan.



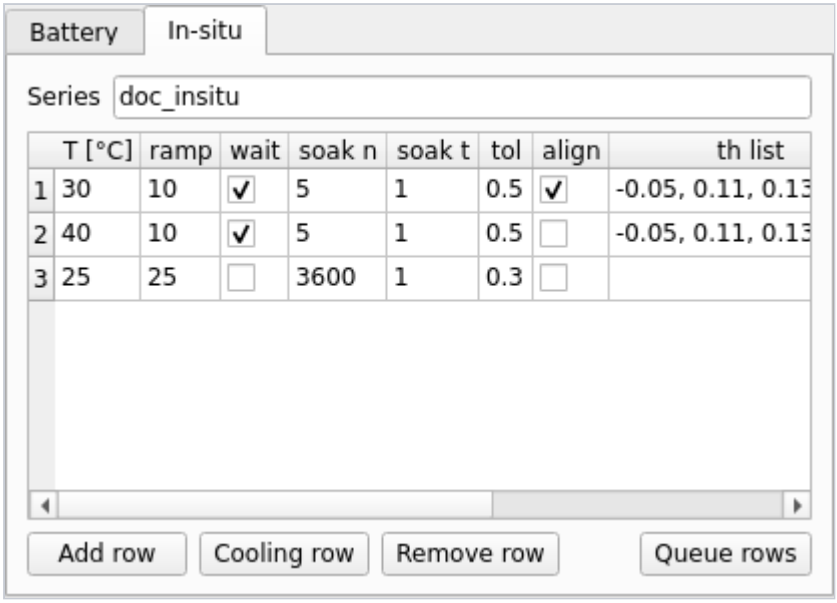
A finished scan with its statistics

When the scan ends, the line above the plot gives CEN, FWHM, COM and the maximum and its position, and markers are drawn: CEN red, COM green, MAX blue, and the stage's target black with the pick's name. In the picture the stage picked CEN, so the black target lies on the red CEN line. The line and the coloured markers are of the points as taken; where the outlier filter replaced a point, the target is of the filtered points and can lie off them. A scan that did not end in success shows its exit status and why; the whole line is in its tooltip when it does not fit.

The **Scan** list keeps this window's last 100 scans; pick one to look at it again. Every scan, from any window or none, is also appended to the SPEC-format file of its item's data folder, `<series>.spec` (see Data and install.md section 4.5), for splot or PyMCA. The toolbar is silx's: zoom, pan, log axes, grid, save, print.

In-situ (film against temperature)

The In-situ form is SPEC's `myinsitu` as a table. Each row is one `insitu_step`; the rows run in order and their frames go to one series.



The in-situ table: 30 °C, 40 °C, then the cooling row

Column	Default	What it is
T [°C]	empty	the setpoint (`settemp T`); empty leaves the temperature alone
ramp	empty	°C/min (`set_ramp`)
wait	off	wait until T is reached (`changec T` instead of `settemp T`)
soak n	empty	`loopscan` points while soaking
soak t	empty	seconds per soak point
tol	0.3	with wait: how close to T counts as reached, °C (`changec`'s `DELTA_K`)
align	on	run `align_in.mac` (the Thin film alignment) before the frames
th list	-0.05, 0.11, 0.13, 0.15	one frame at each th, comma separated; empty takes no frames
texpo	-70000	seconds (> 0) or monitor counts (< 0)
name	empty	the frames' file name

Series names the data folder and stack of every row; empty means the first row's name.

One row runs, in order:

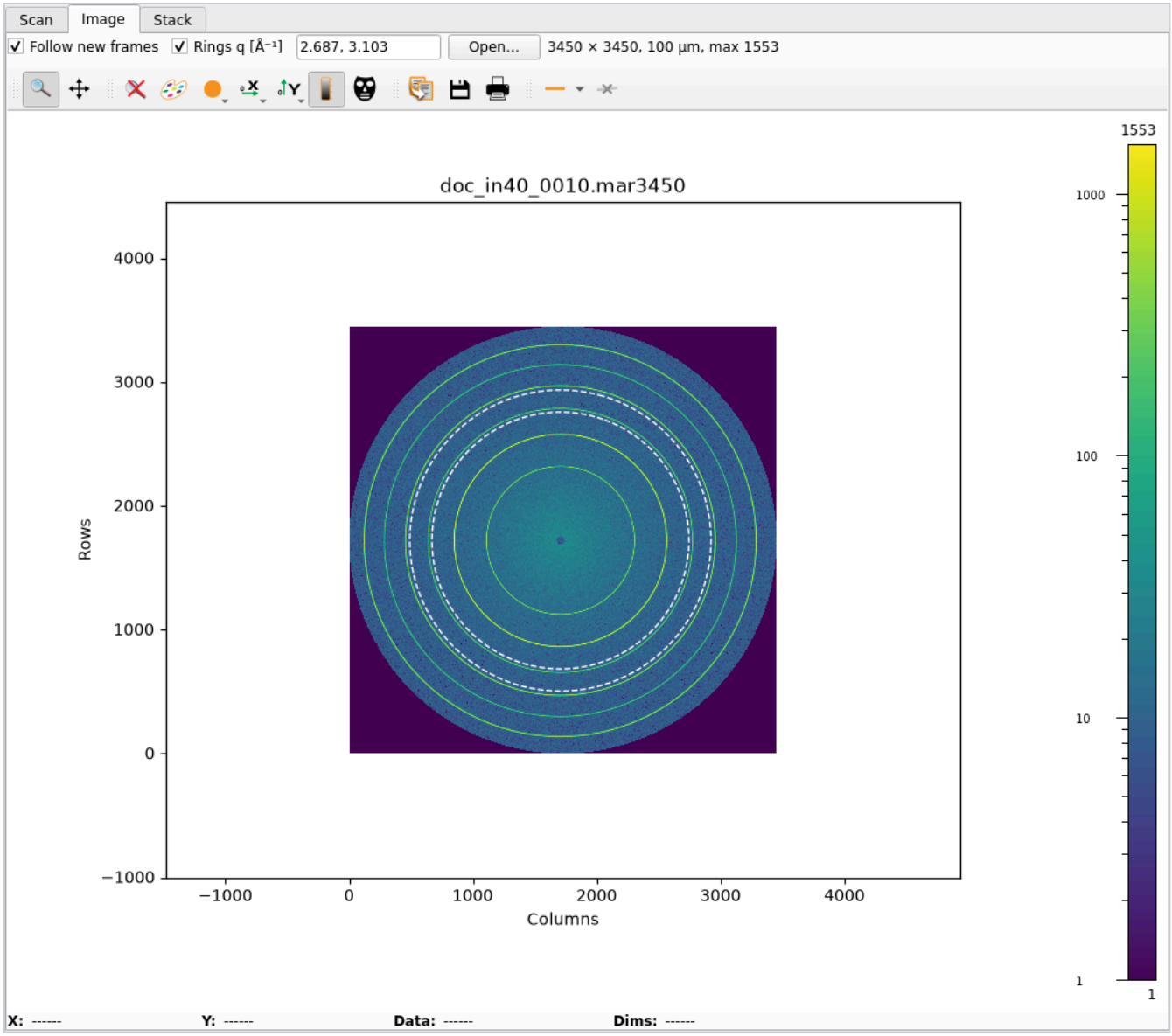
1. Temperature, only the parts filled in: `qdo` of the beamline's `temperature.mac`, `set_ramp <ramp>`, then `settemp T`, or with wait `changec T` with `DELTA_K` set to tol, then `loopscan n t`.
2. With align: the Thin film alignment above.
3. With a th list: `mymar_in`. If tth is below 40 it first closes the shutter and moves through `uan 45 0` to the in-situ positions. Then `uan 81 0`, `counters 2 3`, `fi 1 0 0 0`, `ic_range 5`, `ic_offset 0`, `mar_ip7 25 440 42.4 340 81.8 1.2 -6.8`, 5 s, `scell` to 0.9, a frame at each th, `ic_offset 1`, `fi 1 0 1 1`, back to tth 42 with the mar345 out of the way, and `uan 1 0`.

Button	What it does
Add row	an empty row with the defaults
Cooling row	`myinsitu`'s last step: ramp 25 °C/min to 25 °C, no wait, soak 3600 points of 1 s, no alignment, no frames
Remove row	removes the selected rows
Queue rows	checks every row, then queues them all in order

The checks: name 1 to 27 characters without spaces or quotes (only when the th list is not empty); texpo not 0; ramp above 0; wait needs a T; with wait, tol above 0 (without wait tol is not used); soak needs both n (a whole number, at least 1) and t (above 0); a row with no T, ramp or soak, no alignment and no th list does nothing and is refused. A bad row opens an "In-situ" box that names the row and the field; nothing is queued then.

The cooling row soaks for an hour. Queue it last.

Image tab

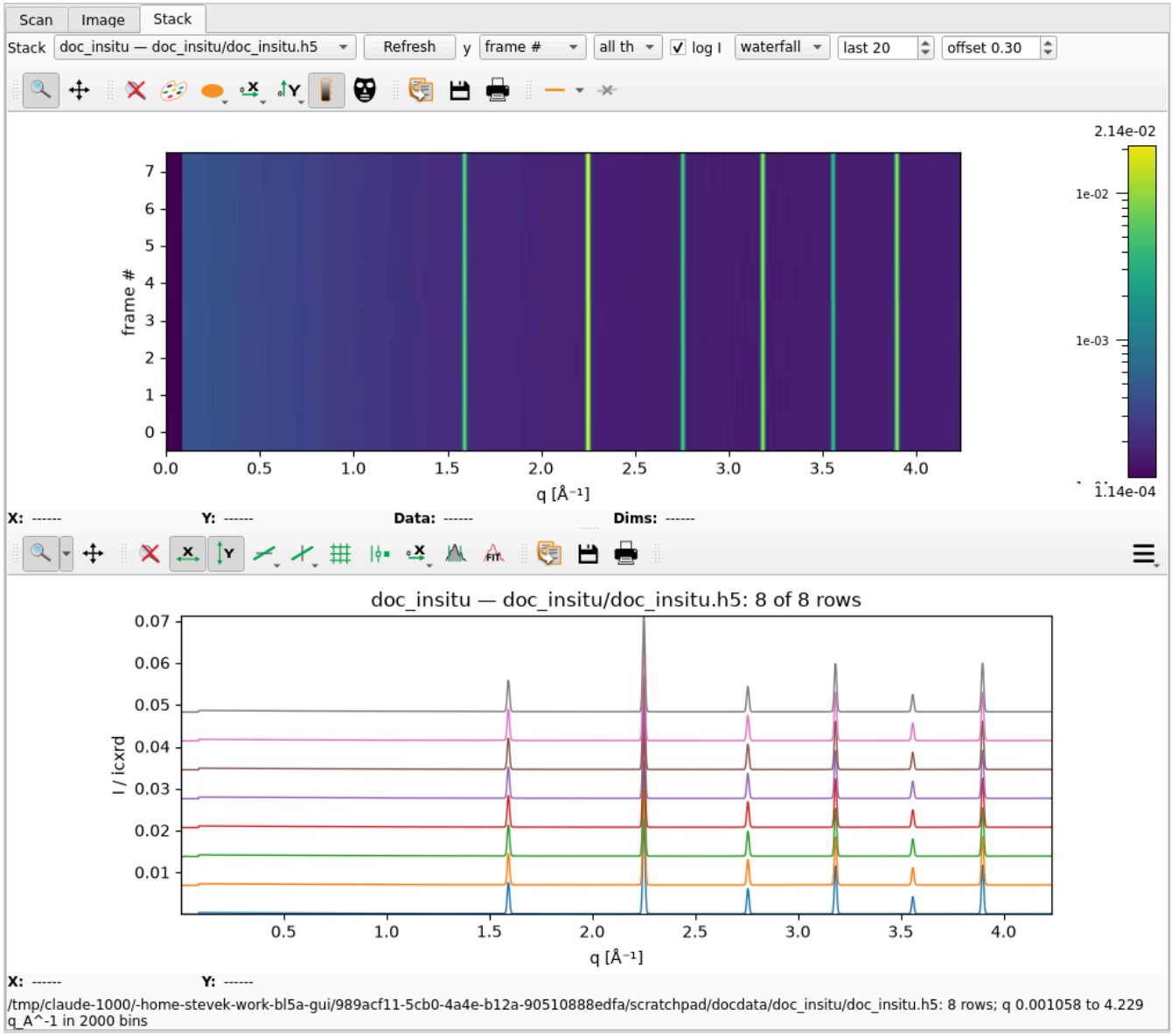


The last frame with the reference rings

- **Follow new frames**: each new frame is loaded as it is written. The file is read off the GUI's thread, from the path the mar345 IOC reports, so that path must be readable on the GUI's computer.
- The image is drawn on a log colour scale; zero counts are drawn as 1.
- **Rings q [Å⁻¹]**: comma-separated q values, drawn as dashed white circles where the integrator's geometry puts them. The defaults are Al 111 and Al 200 (2.687 and 3.103). The simulated film in the picture has no aluminium, so its rings fall elsewhere.
- **Open...** loads any mar345 file.
- The line after the buttons gives the image size, the pixel size and the largest count, or the reason a file could not be read.

The rings use the integrator's PONI, which the GUI asks the integrator for. The integrator uses one PONI for every frame, so the rings and the stacks are right only for the detector position that PONI was measured at (see install.md).

Stack tab



A series stacked: the heatmap and the waterfall

The integrator integrates every frame and appends it to its series' stack in the item's data folder. A stack is one series on one q axis (see Integration). The tab shows one stack:

- the heatmap: q across, one row per frame (or by time), log colour;
- the profiles: I / icxrd against q for the last N rows.

Control	What it does
Stack	every stack the integrator holds, as `series — folder/file`, in path order; at start the last one is shown
Refresh	asks the integrator again for its settings and its stacks
y	`frame #`, or `time [min]` from the first frame
all th / th v	all frames, or only those taken at one th
log I	log colour scale (default on)
overlay / waterfall	profiles drawn on top of each other, or stepped up

Control	What it does
last N	how many of the newest rows the profiles show (default 20)
offset	the waterfall step, as a fraction of the rows' median maximum (default 0.3)

New rows arrive by themselves. A new stack shows up in the list when its first frame is integrated; the tab stays on the stack shown, or shows the new one when none was shown.

The line at the bottom gives the stack's file, its number of rows and its q axis. It reads "integrator: not answering" when the integrator does not reply. The integrator's settings are in the Integration box.

The integrator reads a folder's stacks once, at its start or at the folder's first row. A stack file deleted on disk while it runs stays in the list until the series' next frame, which starts the file again.

SPEC console, Commands, Messages



The SPEC console

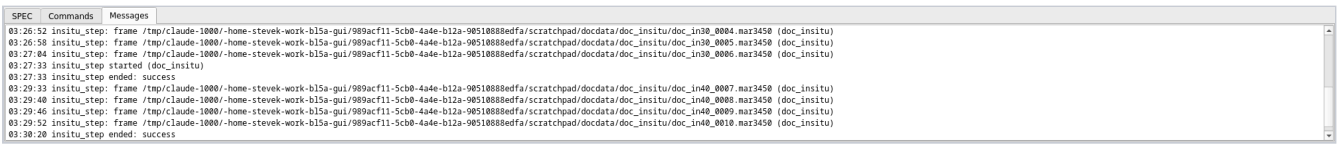
SPEC shows what SPEC prints, as spec-ioc's `Log` PV carries it, the plans' output included. `--- N lines not shown ---` marks a burst longer than the PV holds between two updates; `--- spec-ioc restarted ---` marks an IOC restart.

The entry sends one SPEC command (Enter or **Send**) when the queue is idle (see Who drives SPEC). The command is echoed as `HH:MM:SS SPEC> command`, followed by SPEC's reply. A command that does not end OK adds a line with its status. **^C** interrupts the command sent from here; it never touches a plan's command.



The commands the plans sent

Commands lists every SPEC command a plan sent, with the time and the plan, and its reply indented below.



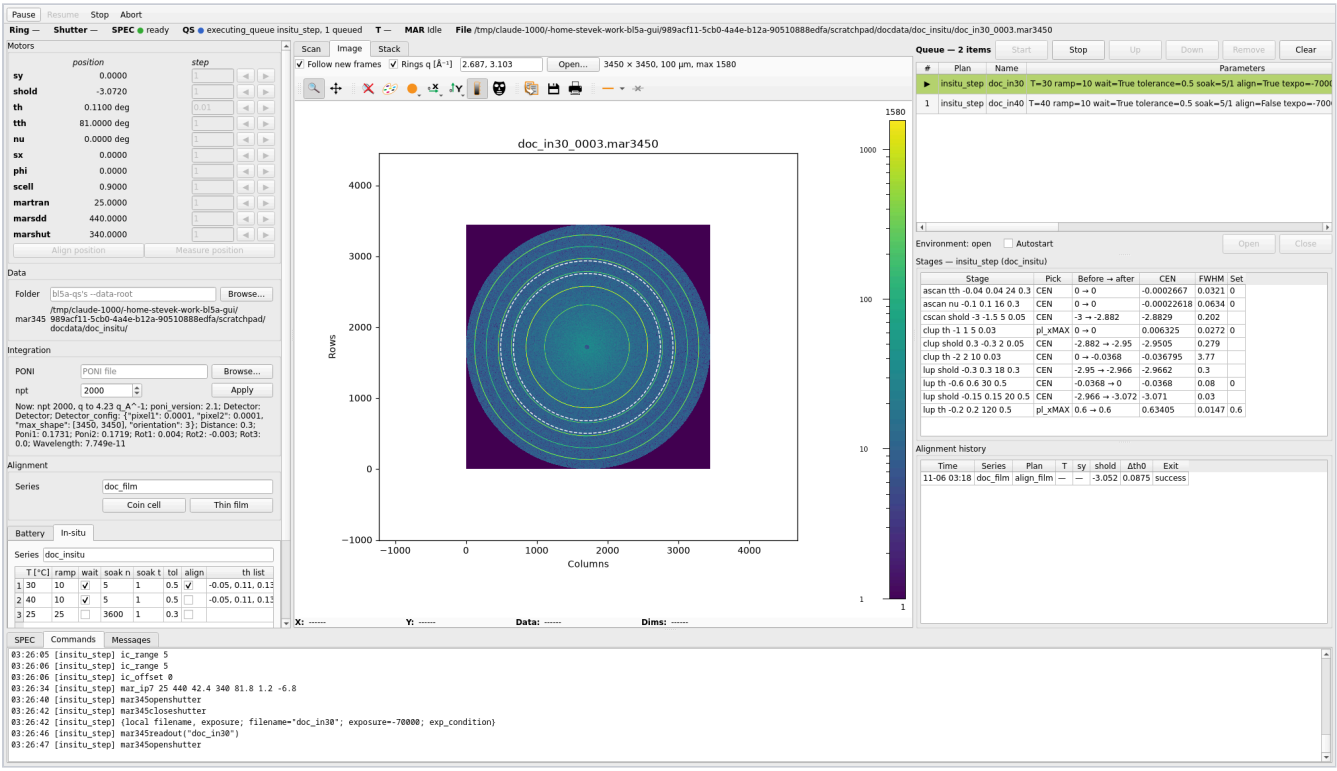
The GUI's messages

Messages is the GUI's own log: each queue request and its result, each plan's start, each queue item's end with its exit status (and, when it did not succeed, why), each frame with its series, and errors from bl5a-qs and the integrator. A request bl5a-qs refuses also opens a "bl5a-qs" box.

Walkthrough: a battery cycle

1. Mount the coin cell. Check the **Data** folder. In **Alignment**, type a series (for example the cell's name) and click **Coin cell**.
2. Click **Start** in the queue. The Scan tab draws the sy and shold fly scans; the history gets a row with the sy and shold the run left.
3. With the queue idle, type `umv tth 81` in the SPEC console.
4. On the **Battery** form: a file name, Texpo, **Repeat** with N and Ti (or **Single shot**), the geometry and the same series, so the frames go to the alignment's folder. **Add to queue**, answer **Yes** to the beamstop question, then **Start**.
5. Start the cycler from its own computer.
6. Each frame shows on the **Image** tab and is added to the **Stack** tab; choose `time [min]` for y to follow the cycle.
7. To end before N frames, click **Stop after frame**.

Walkthrough: a film against temperature



An in-situ run: the frame, the queue, the stages and the commands

1. Mount the film. Check the **Data** folder and, for the detector position `mymar_in` uses, the **Integration** PONI. On the **In-situ** form, type the series.
2. Fill one row per temperature: T, ramp, wait and tol, a soak if wanted, align, the th list, texpo and the file name. Add the **Cooling row** last.
3. **Queue rows**, then **Start**.
4. Each row sets the temperature, aligns (Stages and Scan tabs) and takes its frames (Image and Stack tabs). The history gets a row per aligned step with its T.
5. On the **Stack** tab, pick one th (for example `th 0.13`) to compare the same angle across temperatures.

AI control (MCP)

`gui/run_mcp.py` gives Claude Code or Codex the GUI's controls as MCP tools (setup: install.md). The client starts it through `deploy/bl5a-mcp` when it starts in the `bl5a_gui` directory. It builds queue items with the forms' own code, passes the same gate as the buttons and writes only the PVs the GUI writes, so an item the AI queues is the item the form would queue. It runs beside the GUI, and the GUI shows what the AI queued and runs.

Ask in plain words, for example "queue a thin-film alignment for series film3 and start it", "show me the last frame" or "what did the last fly scan find?".

Tools	What they do	In the GUI
`status`	the queue's and SPEC's state, and `allowed`: the actions open now	status line; which buttons are live
`queue`, `history`	the waiting items; the finished ones with their exit status	Queue
`motors`	position, step and unit of each motor	Motors
`scans`, `scan`, `alignment`	the scans with their peak values; the stages and the alignment history	Scan tab, Stages, Alignment history
`messages`, `spec_log`, `commands`	the three logs	Messages, SPEC, Commands
`frame_image`	the last frame, or a named one, as a picture	Image tab
`integration`, `stacks`, `stack`	the integrator's PONI and npt; the stacks; one stack's rows	Integration, Stack tab
`queue_alignment`, `queue_mar_series`, `queue_insitu`	queue items	Alignment buttons, Battery form, In-situ form
`queue_remove`, `queue_move`, `queue_clear`, `queue_start`, `queue_stop`, `queue_autostart`, `environment_open`, `environment_close`	edit and run the queue	the queue's buttons
`pause`, `resume`, `stop`, `abort`, `stop_after_frame`	end or hold the running plan	Run buttons, Stop after frame
`tweak`	move a motor by a step	Motors ◀ ▶
`spec_command`, `spec_result`, `spec_abort`	one SPEC command, its result, ^C	SPEC console
`integration_set`	a PONI file and npt for the next frames	Integration Apply

The rules are the GUI's:

- An action that is not open now is refused with the reason, where the button would be grey: `pause: not open now (qs idle; …)`. The tools ask bl5a-qs first, so each one sees what the previous one did.

- `data_folder` left out uses the GUI's Data folder; `""` uses bl5a-qs's `--data-root`; an absolute path is used as it is. The data then goes to `<folder>/<series>`.
- `queue_mar_series` needs `beamstop_in_place: true`, the GUI's beamstop question. The server tells the AI to ask you first, and to ask before `abort` and `queue_clear`. Your MCP client's permission prompt is the other guard: allow only the read tools without a prompt.
- `pause` then `resume` fails the plan, as in the GUI. `stop_after_frame` ends a series after the frame being taken; asked before the first frame starts, the series stops with none.
- `spec_command` waits up to `wait_s` (default 30 s); a longer command comes back as running, and `spec_result` gives its end. What SPEC prints (the table of `wm`, for example) goes to the SPEC log, not the reply: read it with `spec_log` from the `spec_log_since` the result gives.
- The logs hold what happened since the server started. Each read gives `last`; pass it as `since` next time to get only the new lines.

As on the Battery form, move tth to 80° or more before a standard `queue_mar_series` (Battery measurements); a thin-film alignment leaves it near 1.2. Use `tweak`, or `spec_command` with `umv tth 81` at the beamline.

When something is wrong

What you see	What it means	What to do
QS ● not answering	bl5a-qs does not reply on `--qs-control`	`systemctl --user status bl5a-qs`, its journal (`journalctl --user -u bl5a-qs`), or the address
SPEC —	spec-ioc's PVs do not connect	`systemctl --user status bl5a-spec-ioc`; check Channel Access to its host
SPEC ● disconnected	spec-ioc runs but SPEC's server does not answer it	start SPEC with `-S`; check the host and port in `SPEC=`
Motor buttons and the console stay grey	a plan runs, SPEC is busy, or a console command runs	wait, or stop the queue
"Battery", "In-situ" or "Alignment" box	a field breaks a rule, a series that is not a folder name included	the box names the field and the rule
"bl5a-qs" box	bl5a-qs refused a request	the box gives its reason
Messages: "... ended: fail: ..." and Autostart unticked	the item failed; the text after `fail:` says why	the last SPEC command and its reply are at the bottom of Commands
Image info line shows a file error	the frame's path is not readable here	mount the mar345 data directory at the same path on this computer
Stack: "integrator: not answering"	bl5a-integrator is not running or not at `--integrator-rep`	`systemctl --user status bl5a-integrator`; Refresh
"Data" box	the Data folder is not an absolute path, holds a space, or with the series is over 200 bytes	the box gives the rule
Data: mar345 folder "missing"	the mar345 IOC does not see the item's folder	mount the data root at the same path on the IOC's host
Messages: "... ended: fail: device state error: data folder: ..."	the data root does not exist on bl5a-qs's host	create it, or fix the Data folder and queue again

What you see	What it means	What to do
Integration: "Refused: …"	the integrator could not use the PONI or npt	the text after "Refused:" says why
T reads "—"	no temperature PVs are configured	the temperature is set by SPEC's macros; read it in SPEC